

# A recursive pathway for isoleucine biosynthesis arises from enzyme promiscuity

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## eLife Assessment

The study reports a potential pathway for isoleucine biosynthesis mediated by the underground activity of AHASII, which converts glyoxylate and pyruvate to 2-ketobutyrate. While the findings are **valuable** in revealing a possible alternative route for isoleucine production, the evidence presented remains **incomplete**. More comprehensive biochemical experiments are required to substantiate the physiological feasibility of this pathway.

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## Abstract

Enzyme promiscuity can be the starting point for the evolution of new enzymatic activities and pathways. Previously [Cotton et al. \(2020\)](#) identified underground isoleucine biosynthesis routes that can replace the canonical route in *Escherichia coli*, after they deleted the enzymes that catalyze the formation of its precursor 2-ketobutyrate. Using this strain and short-term evolution we identify a new pathway for isoleucine biosynthesis based on the promiscuous activity of the native enzyme acetohydroxyacid synthase II. We demonstrate that this enzyme catalyzes the previously unreported condensation of glyoxylate with pyruvate to generate 2-ketobutyrate in vivo. The gene encoding this enzyme, *ilvG*, is inactivated by a frameshift mutation in the laboratory model strain *E. coli* K-12 MG1655. Its evolutionary reactivation we report here points to a potential natural role in isoleucine biosynthesis in *E. coli*. Isoleucine biosynthesis proceeds with a further condensation step of 2-ketobutyrate with pyruvate, again catalyzed by AHAS, giving the proposed pathway the unusual property of recursivity. The discovered enzyme activity uses glyoxylate and pyruvate as direct central metabolic precursors for isoleucine biosynthesis instead of its canonical ‘indirect’ biosynthesis via the amino acid threonine. Unlike previously discovered underground isoleucine routes by Cotton et al., this route is more likely to play a role in natural isoleucine biosynthesis in *E. coli* due to the use of ubiquitous metabolites and its activity in aerobic conditions. The discovered route further expands the known metabolic space for isoleucine biosynthesis in *E. coli* and

potentially other organisms, and could find applications in biotechnological isoleucine production.

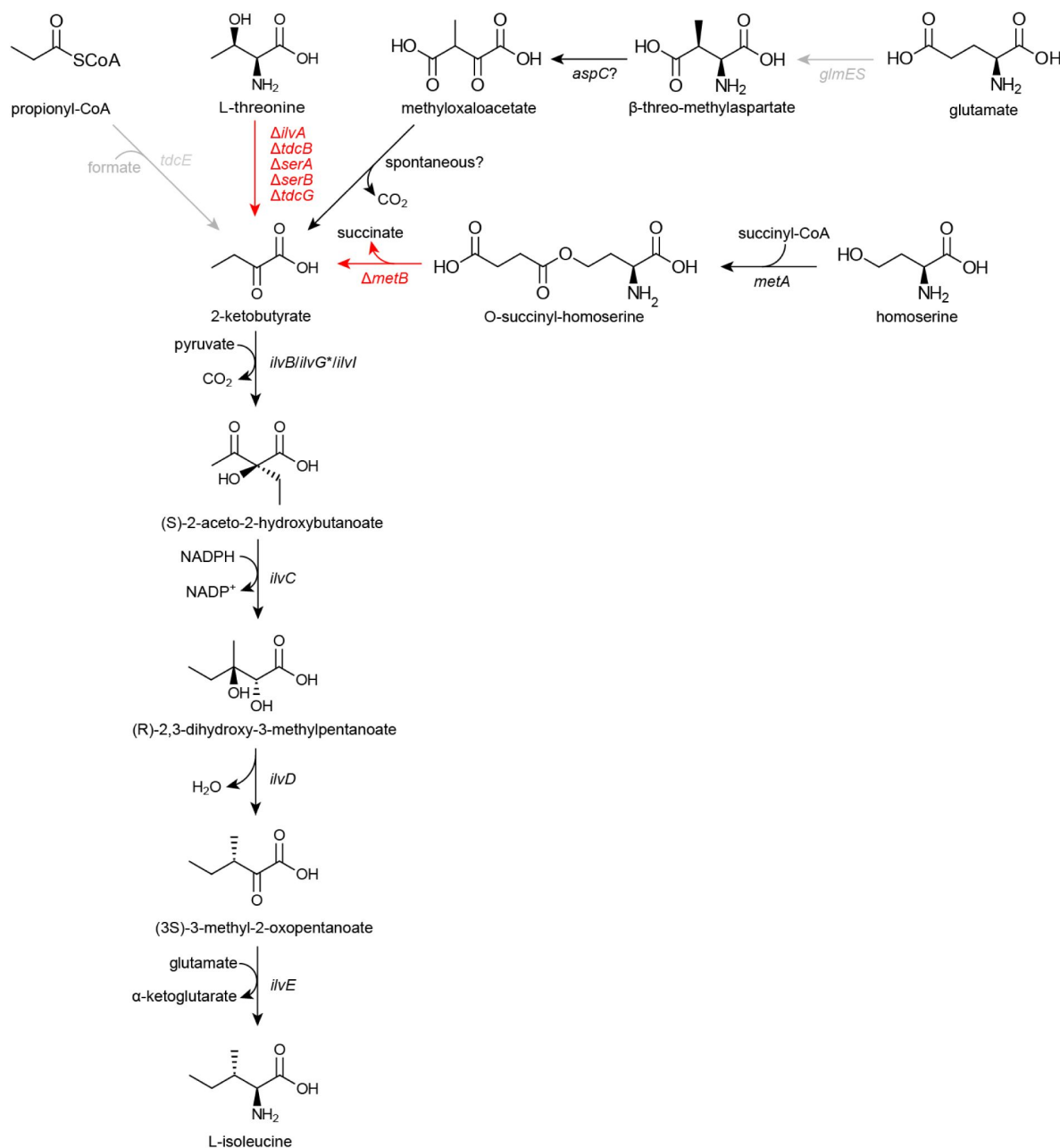
## Introduction

Central metabolism includes all the reactions that are necessary to produce building blocks for essential biomass components, such as amino acids for the synthesis of proteins. Amino acid biosynthesis pathways have been the subject of intense scrutiny over many decades to elucidate the biochemical and physiological details of each component of the network (Neidhardt, 1987 [↗](#); Noor et al., 2010 [↗](#); Umbarger, 1978 [↗](#)). Despite these efforts, our understanding is incomplete on multiple levels, even for model organisms such as *Escherichia coli*. This is partially explained by the fact that enzymes are often able to accept multiple substrates, a property called promiscuity, leading to overlap and crosstalk between pathways (Noda-Garcia et al., 2018 [↗](#)). Promiscuity increases the complexity of metabolism, but it is beneficial in terms of network stability and resilience. Additionally, promiscuous activities can be the starting point for the evolution of new enzymatic activities and pathways (D'Ari & Casadesús, 1998 [↗](#); Khersonsky & Tawfik, 2010 [↗](#)).

An emerging method to study enzyme promiscuity and underground metabolism is the use of microbial strains in which auxotrophies for certain key metabolites are created via one or multiple genetic deletions (Kim et al., 2010 [↗](#); Pontrelli et al., 2018 [↗](#); Wenk et al., 2022 [↗](#)). Such synthetic auxotrophic strains are often also referred to as biosensors strains or growth-coupled selection strains, as they can 'sense' the presence of a certain metabolite or essential building block, with growth as a simple experimental readout. Especially for *E. coli* a wide collection of sensor strains has been built in recent years (Schulz-Mirbach et al., 2025 [↗](#)). Metabolites for which these strains are made auxotrophic can be supplied externally in the growth medium to support their growth. When the metabolite is left out of the medium, these strains can be tested with new synthetic routes for the biosynthesis the metabolite, and this method is increasingly used in metabolic engineering of synthetic metabolic routes (Orsi et al., 2021 [↗](#)). Alternatively, these sensor strains can be used to identify and potentially evolve latent native routes for biosynthesis of essential metabolites. This approach using synthetic auxotrophic strains has for example led to the identification of novel routes for the essential cofactors coenzyme A and pyridoxal-5'-phosphate biosynthesis in *E. coli* (Kim et al., 2010 [↗](#); Pontrelli et al., 2018 [↗](#)).

Recently, an engineered isoleucine auxotroph *E. coli* strain was reported in which all the threonine deaminases were deleted, leading to a lack of the threonine-derived isoleucine precursor 2-ketobutyrate (2KB) (Cotton et al., 2020 [↗](#)). Using this strain the authors reported two previously unknown routes for 2KB biosynthesis. The first route is based on the promiscuous O-succinyl-homoserine  $\gamma$ -cleavage of the native cystathionine  $\gamma$ -synthase encoded by *metB*, an essential enzyme in methionine biosynthesis. Another is based on the condensation of propionyl-CoA and formate by 2KB-formate lyase TdcE. Their findings brought the number of known 2KB biosynthesis pathways in nature to nine, of which four have been demonstrated in *E. coli* (Abramsky et al., 1962 [↗](#); Cotton et al., 2020 [↗](#)) (Figure 1 [↗](#)).

Using a similar isoleucine/2KB auxotrophic *E. coli* strain, we discovered that a key enzyme downstream of 2KB in isoleucine biosynthesis, acetohydroxyacid synthase (AHAS) II (encoded by *ilvG*), also can support the biosynthesis of 2KB. We report that AHAS II is able to accept the smaller 2-keto acid glyoxylate in addition to its canonical substrates pyruvate and 2KB. The condensation of glyoxylate with pyruvate eventually generates 2KB, giving rise to a previously unknown recursive pathway for isoleucine biosynthesis, which we demonstrate with isotopic labeling to elucidate metabolic fluxes. This is the tenth discovered pathway for isoleucine in nature, and provides further insights in the metabolic space and promiscuous networks of amino acid biosynthesis. This shorter isoleucine biosynthesis route may be widespread in nature and can be a potentially attractive pathway for isoleucine production in biotechnology.



**Figure 1.**

**Scheme of the isoleucine/methionine auxotroph strain (IMaux) used in this study as a 2KB biosensor, showing the four previously identified isoleucine biosynthesis pathways in *E. coli*.**

Red arrows indicate gene deletions, grey arrows indicate reactions that are inactive in the presence of oxygen (in addition, *glmES* is present in some *E. coli* strains but not in *E. coli* MG1655). *ilvG\** indicates that this gene is truncated in *E. coli* MG1655 due to a frameshift and hence inactive

## Results

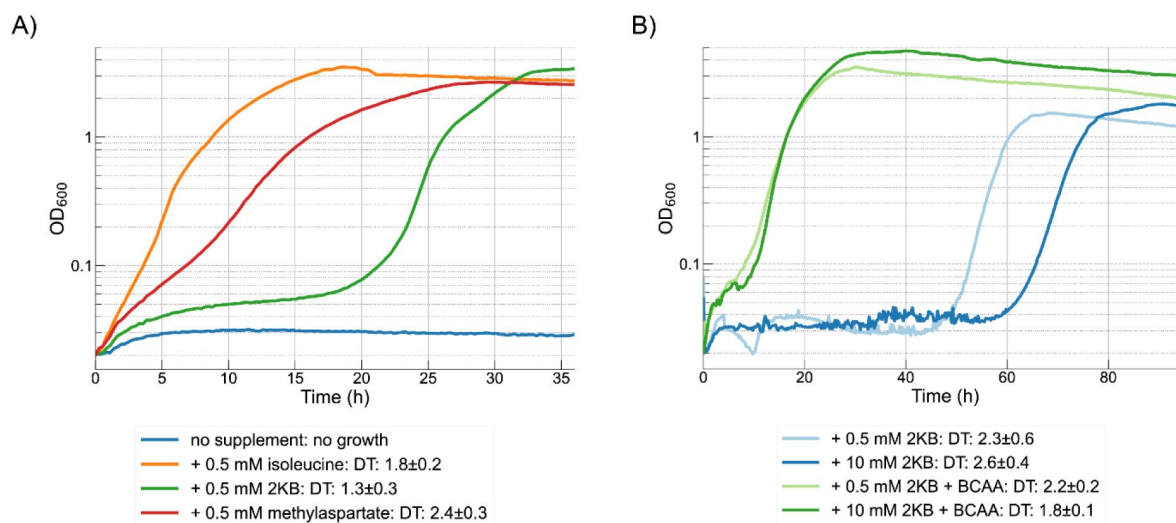
### Characterization of an *E. coli* 2KB biosensor strain

We obtained the  $\Delta 5$  isoleucine auxotroph strain (SIJ488  $\Delta ilvA \Delta tdcB \Delta sdaA \Delta sdaB \Delta tdcG::kan$ ) as a kind gift by Dr. Charlie Cotton with the intention of testing alternative 2KB biosynthesis routes (**Figure S1** [↗](#)). For this reason, we additionally deleted cystathionine  $\gamma$ -synthase ( $\Delta metB$ ), since it was reported that this native enzyme involved in methionine biosynthesis, could also promiscuously support 2KB biosynthesis via O-succinylhomoserine cleavage activity (**Figure 1** [↗](#)) (Cotton et al., 2020 [↗](#)). This deletion in the methionine biosynthesis pathway results in a methionine auxotrophy, so we named the resulting strain isoleucine-methionine auxotroph (IMaux). Another known pathway for 2KB biosynthesis in *E. coli* relies on the reverse activity of 2KB formate lyase (encoded by *tdcE*), however this enzyme requires a glycyl radical and can only operate under anaerobic conditions, and hence its deletion was not required as we only worked under aerobic conditions (**Figure 1** [↗](#)). A last known pathway, via methylaspartate, also did not require a deletion, as this metabolite does not commonly occur in *E. coli* since it requires a glutamate mutase which is not present in *E. coli* MG1655, though it is present in other strains of *E. coli* such as the pathogenic strain O157:H7, where its native role is expected to be glutamate fermentation (Kronen & Berg, 2015 [↗](#)), but it may also play a role in isoleucine biosynthesis (Abramsky et al., 1962 [↗](#); Phillips et al., 1972 [↗](#)). Additionally, glutamate mutase is an oxygen-sensitive enzyme and it requires the coenzyme B<sub>12</sub>, which *E. coli* cannot synthesize (Fang et al., 2018 [↗](#)) and is not present in the M9 minimal medium used in this study.

The IMaux strain cannot grow on glucose or glycerol unless both methionine and isoleucine or 2KB are provided (**Figure 2A** [↗](#)), confirming the functionality of the sensor strain. Notably, the strain displays a lag phase when supplemented with as little as 0.5 mM of 2KB (**Figure 2A** [↗](#)), suggesting toxicity of this compound to *E. coli*. The mechanism of this toxicity is unknown, though several hypotheses have been put forward: interference with native metabolism via competitive inhibition (Bisswanger, 1981 [↗](#)) due to the structural similarity between 2KB and pyruvate, the role of 2KB as a so-called ‘alarmone’, signaling the shift between aerobic and anaerobic growth (Daniel et al., 1983 [↗](#)), and the inhibition of acetolactate synthase (ALS) activity, leading to valine and/or leucine starvation (LaRossa et al., 1987 [↗](#)). We reasoned that if the inhibition of ALS activity is responsible for the toxicity effect, supplementation with branched-chain amino acids (BCAA) valine, leucine, and isoleucine should remove the toxicity. Our results indeed demonstrate that 2KB toxicity is mediated by dysregulation in BCAA metabolism, since supplementation removes the lag phase (**Figure 2B** [↗](#)).

### Confirming the methylaspartate route for isoleucine biosynthesis in *E. coli*

Next, we tested the strain’s performance as a 2KB biosensor by supplementing methylaspartate, the intermediate that in some *E. coli* strains under anaerobic conditions can be formed from glutamate by glutamate mutase. The formation of 2KB from methylaspartate has been reported in *E. coli* strains W and Crookes (Abramsky et al., 1962 [↗](#); Phillips et al., 1972 [↗](#)). This pathway presumably proceeds through transamination to methyloxaloacetate via the activity of a promiscuous transaminase (most likely aspartate transaminase, AspC), though this activity has not previously been confirmed to the best of our knowledge. The product of this transamination, methyloxaloacetate, is a beta-keto acid which can undergo spontaneous decarboxylation yielding 2KB as a product (Kubala & Martell, 1981 [↗](#)), similarly to what is observed for oxaloacetate and pyruvate (Tsai, 1967 [↗](#)). Our results show that supplementation of 0.5 mM of methylaspartate fully



**Figure 2.**

### IMaux strain characterization.

(A) The strain can serve as a sensor for isoleucine, 2KB, or compounds that can be converted to 2KB. Methylaspartate can be converted to 2KB in vivo via promiscuous reactions of native *E. coli* enzymes (B) Supplementation of branched chain amino acids valine, leucine, and isoleucine (BCAA) at 1 mM each relieves 2KB toxicity, indicating that the mechanism involves dysregulation of BCAA biosynthesis. All experiments have been performed in M9 minimal medium with 20 mM glycerol as a main carbon source and 1 mM methionine. Growth curves are the average of at least three technical replicates. DT, doubling time.

restores the strain's growth (**Figure 2A** [↗](#)), indicating that this sequence of reactions is indeed taking place, and that methylaspartate can be efficiently imported, presumably through one of the native dicarboxylate transporters such as DctA ([Uندن et al., 2016](#) [↗](#)).

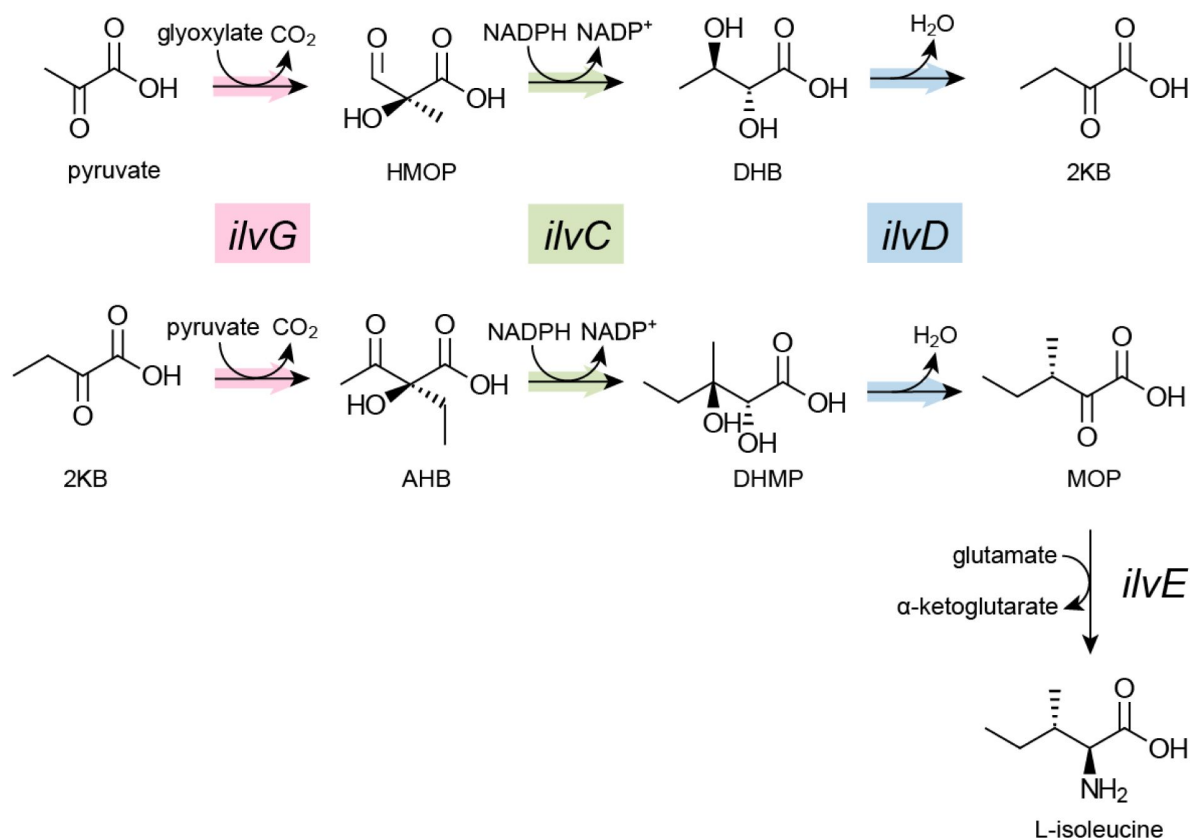
## Discovery of a novel, recursive isoleucine biosynthesis pathway in the 2KB biosensor

We initially intended to test a hypothetical 2KB biosynthesis pathway that requires the condensation of propionyl-CoA and glyoxylate (**Figure S1A** [↗](#)). For this reason, we performed negative control experiments to test whether the IMaux strain could grow when supplemented with glyoxylate and/or propionate. Unexpectedly, we noticed growth after a lag phase of about 100h when 10 mM glyoxylate was supplemented (data not shown), potentially indicating that this compound can be converted to 2KB via an unknown mechanism.

We isolated and sequenced the resulting strain to confirm the presence and nature of genomic mutations. The only mutation we found was an insertion of two nucleotides in the *ilvG* pseudogene (+AA at position 981) reverting the frameshift that is normally present in the laboratory strains of *E. coli* such as the K-12 derivative MG1655 strain used as a base for these experiments ([Gray et al., 1981](#) [↗](#)). This mutation has already been reported, and it is known to lead to the production of the enzyme acetohydroxyacid synthase (AHAS) II ([Lawther et al., 1981](#) [↗](#), [1982](#) [↗](#)). The AHAS II enzyme is one of the three AHAS complexes found in *E. coli*. The AHAS complexes I, II, and III, are encoded respectively by *ilvBN*, *ilvGM*, and *ilvIH*, respectively ([Eoyang & Silverman, 1984](#) [↗](#); [Gollop et al., 1989](#) [↗](#); [Hill et al., 1997](#) [↗](#)). Each of these is composed of a larger, catalytic subunit (IlvB, IlvG and IlvI) and a smaller, regulatory subunit (IlvN, IlvM or IlvH). These enzymes were already known for a function in the biosynthesis of isoleucine, but this role is downstream of 2KB biosynthesis, as they condense 2KB with pyruvate into aceto-hydroxybutanoate. This enzyme complex has been extensively studied for this role in vitro, and also the promiscuous activity for the condensation of two pyruvate molecules for valine biosynthesis is known ([Steinmetz et al., 2010](#) [↗](#); [Vyazmensky et al., 2011](#) [↗](#)).

While IlvG is obviously involved in the isoleucine biosynthesis pathway, it was initially unclear why this mutation would confer the ability to relieve an isoleucine auxotrophy. The known promiscuity of AHAS II led us to hypothesize that glyoxylate might also serve as a substrate instead of the similar keto-acid pyruvate. This leads to a proposed pathway in which AHAS enzymes recursively catalyze two steps isoleucine biosynthesis (**Figure 3** [↗](#)).

To verify our hypothesis, we deleted *ilvG* from the strain, which aborts growth on glyoxylate, and confirmed that overexpression of *ilvGM* from an inducible plasmid restores growth. Since *ilvG* is the first gene in the *ilvGMED* operon, a deletion could have disrupted the translation or transcription of the other genes in the transcription unit, making the strain unable to synthesize isoleucine regardless of the supplementation of glyoxylate or 2KB, as *ilvE* and *ilvD* are also essential steps towards isoleucine (**Figure 1** [↗](#)). To exclude this possibility, we tested both glyoxylate- and 2KB- dependent growth. We confirmed that the uninduced strain was unable to use glyoxylate to synthesize 2KB, while still being able to use 2KB to produce isoleucine (**Figure 4A** [↗](#)). Based on this result, we concluded that the *ilvG* deletion does not disrupt transcription or translation of the remaining genes in the operon. Moreover, we can conclude that a strain lacking *ilvG* is unable to synthesize 2KB from glyoxylate, meaning that the other AHAS encoding genes *ilvB* and *ilvI* alone cannot substitute for the promiscuous AHAS II activity that is the subject of this paper. This is also implied by the fact that an *ilvG* frameshift reversal mutation was necessary in the evolved strain, and that neither of the counterparts AHAS I and III that are natively present and constitutively expressed were able to support growth on glyoxylate of the 2KB auxotrophic strain.

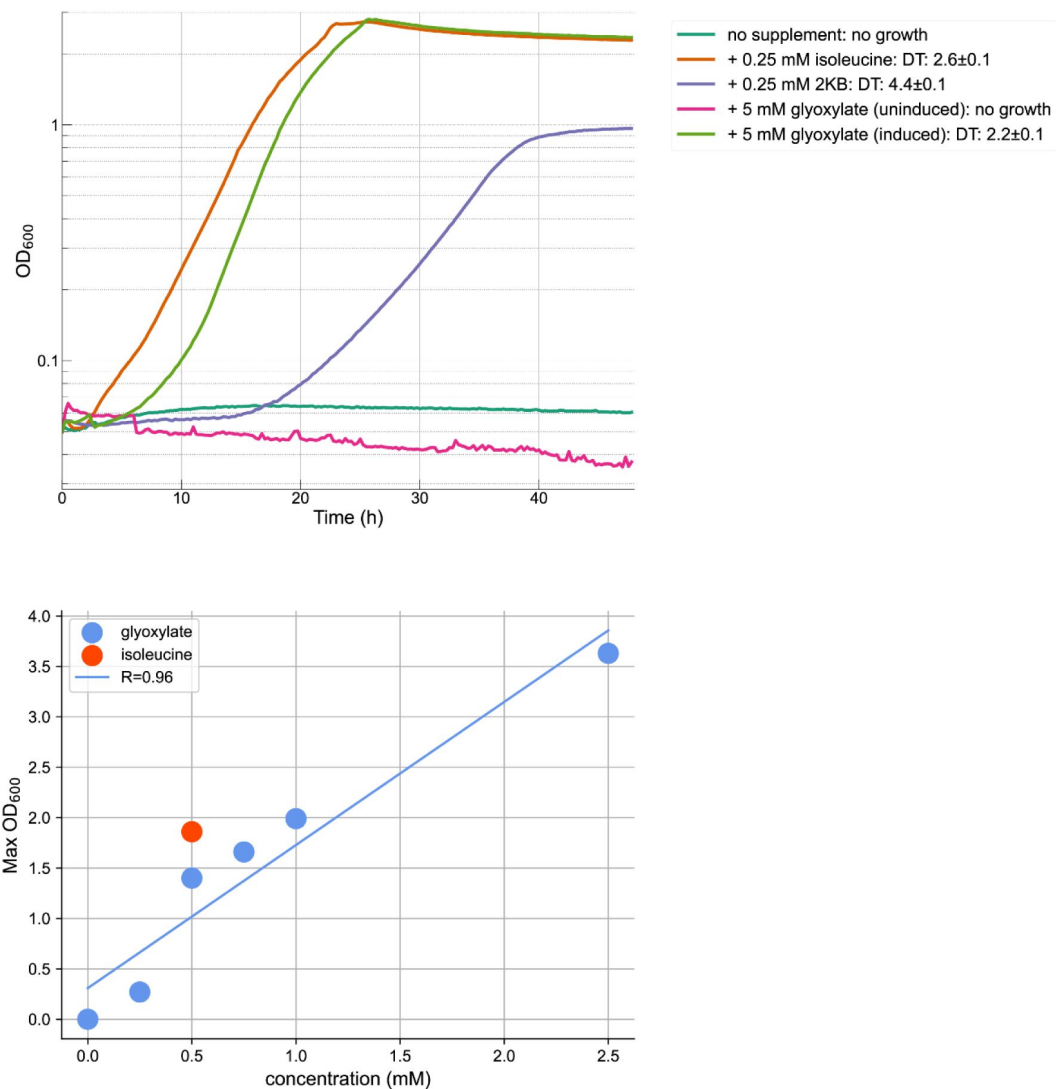


**Figure 3.**

**The proposed recursive isoleucine biosynthesis pathway. Pyruvate is condensed with glyoxylate by AHAS II encoded by *ilvG*, then the product of this reaction is reduced and undergoes dehydration to 2KB.**

The resulting 2KB undergoes the same sequence of reactions to produce the isoleucine precursor (3S)-3-methyl-2-oxopentanoate (MOP), which is further transaminated to isoleucine. AHAS II is the only AHAS enzyme in *E. coli* that can catalyze the first reaction with glyoxylate and pyruvate in vivo, though the condensation of 2-KB with pyruvate can also be mediated by AHAS I and AHAS III. Colored arrows indicate that the same enzyme promiscuously catalyzes both reactions. HMOP, 2-hydroxy-2-methyl-3-oxopropanoate; DHB, (R)-2,3-dihydroxybutanoate; 2KB, 2-ketobutyrate; AHB, (S)-2-aceto-2-hydroxybutanoate; DHMP, (R)-2,3-dihydroxy-3-methylpentanoate; MOP, (3S)-3-methyl-2-oxopentanoate.





**Figure 4.**

**(A)** Growth of the IMAux  $\Delta ilvG$  strain with or without plasmid-based overexpression of *ilvGM*. All strains were grown in M9 with 20 mM glycerol and 1 mM methionine, plus supplemental carbon sources as indicated in the legend. Growth curves are the average of at least three technical replicates. DT, doubling time. **(B)** Max OD reached with different glyoxylate concentrations. Strains were grown in M9 with 20 mM glycerol, 1 mM methionine, and 10  $\mu$ M cuminic acid as inducer. Blue dots indicate a range of glyoxylate concentrations, orange represents isoleucine (positive control).



As expected, we observed a strict correlation between the quantity of glyoxylate available in the medium and the final biomass density ( $R=0.96$ ) (**Figure 4B** [↗](#)).

Next, we used isotopic labeling to confirm our hypothesis that AHAS II produces 2KB from glyoxylate and pyruvate. Based on our proposed route, we expected that if the strains were grown on fully  $^{13}\text{C}$ -labeled glucose and unlabeled glyoxylate, the fourth carbon (C4) of isoleucine should come from glyoxylate, giving rise to a labeling pattern that is distinct from what arises when supplementing unlabeled 2KB (**Figure 5A** [↗](#)). The measured labeling of isoleucine confirmed our hypothesis. Notably, in the unlabeled glyoxylate condition, not all of the isoleucine is M+4 labeled, indicating that some of the supplied glyoxylate is feeding into central metabolism (**Figure 5B** [↗](#)).

This could happen via the activity of glyoxylate carboligase (Gcl) and the so-called ‘glycerate route’, or via the activity of malate synthase (AceB or GlcB). Both of these routes should be inactive in glucose minimal medium because of catabolite repression, though it is possible that amino acid starvation leads to their activation mediated by ppGpp (Traxler et al., 2008 [↗](#)) or that high glyoxylate concentration supplied in the experiments activate these routes. An intriguing alternative is once again presented by the promiscuous activity of AHAS II: this enzyme could also condense two molecules of glyoxylate to produce tartronate semialdehyde, which could then be reduced to glycerate via IlvC. Glycerate could be assimilated via glycerate kinase (GarK), or undergo dehydration to pyruvate via IlvD.

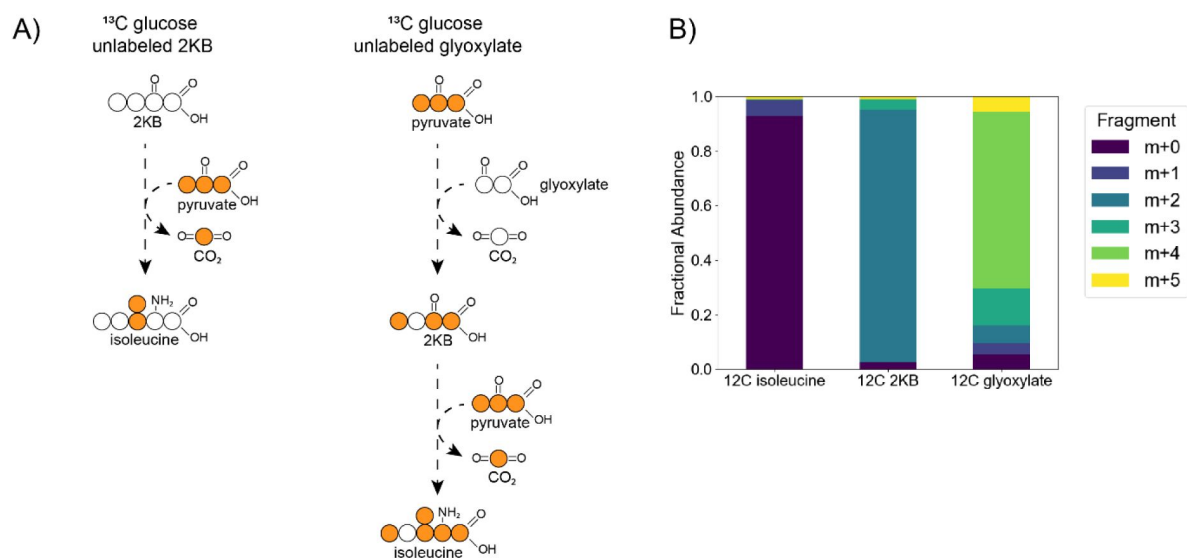
## Discussion

In this study, we identified a previously unknown, recursive pathway for isoleucine biosynthesis dependent on the promiscuous activity of AHAS II. At least nine pathways for 2KB biosynthesis have been characterized in different organisms (Cotton et al., 2020 [↗](#)), however the new route we present here is unique in that it uses an enzyme that is already involved in isoleucine biosynthesis by exploiting a promiscuous activity and producing its own precursor, making the overall pathway recursive, an unusual property in metabolism (Krüsemann et al., 2018 [↗](#); Marcheschi et al., 2012 [↗](#)).

AHAS II is thus able to accept a combination of two-, three-, and four-carbon 2- ketoacids as its substrates, leading to multiple roles in BCAA biosynthesis. AHAS II, and also AHAS I and III, also catalyze an essential step in the canonical biosynthesis route of valine, condensing the 2-ketoacid pyruvate with a second molecule of pyruvate to produce acetolactate (Salmon et al., 2006 [↗](#)).

Furthermore, unlike several of the other isoleucine biosynthesis routes, the one presented in this study is based on a native *E. coli* enzyme and the central metabolites glyoxylate and pyruvate, pointing to a potential natural role, both in *E. coli* and other bacterial lineages. Indeed, the fact that *ilvG* carries a frameshift mutation in lab isolates of *E. coli* such as MG1655 (Lawther et al., 1981 [↗](#)) and BW25113, but not in wild-type isolates from the human gut could point towards the usefulness of this gene in the natural environment. A similar mutation phenotype has been observed in *Salmonella typhimurium* (Burns et al., 1995 [↗](#)), providing further evidence for this hypothesis. How and why such *ilvG* frameshift mutations occurred remains unclear: it has been reported that *E. coli* undergoes cycles of isoleucine starvation in minimal medium due to aberrant metabolic regulation when *ilvG* is not present (Andersen et al., 2001 [↗](#)). However, since lab strains are generally grown and stored on rich LB agar plates, this effect is presumably reduced or eliminated and may relieve the evolutionary pressure to retain *ilvG*.

The novel identified route could also be beneficial under specific growth regimes, e.g. growth on acetate, as during growth on acetate glyoxylate is produced as a key metabolite in the glyoxylate shunt. Interestingly, it has been shown that *ilvG* expression increases when the *E. coli* K-12 strain MC4100, in which *ilvG* is not natively truncated, is grown on acetate (Oh & Liao, 2000 [↗](#)).



**Figure 5.**

**(A)** Expected isoleucine labeling patterns based on <sup>13</sup>C glucose and unlabeled 2KB or unlabeled glyoxylate. Orange and white circles represent labeled and unlabeled carbons, respectively. Dashed arrows indicate multiple reactions. **(B)** Observed isoleucine labeling patterns for strains grown on <sup>13</sup>C glucose supplemented with <sup>12</sup>C carbon sources as indicated in the figure. The GC-MS derivatization method removes the C1 carbon, leaving only the C2-C6 fragment for analysis. Bars represent the average of three technical replicates.

Another potential benefit for the route outside *E. coli* could be in photosynthetic or chemolithoautotrophic organisms that generate glyoxylate during CO<sub>2</sub> fixation using the Calvin cycle. The oxygenase side-activity of the carboxylating enzyme Rubisco, generates phosphoglycolate, which has to be recycled into metabolism via glyoxylate, a key intermediate in all known photorespiration routes (Claassens et al., 2020 [↗](#); Timm et al., 2025 [↗](#)). If the described AHAS II activity is present in such organisms, they could use glyoxylate as a building block for isoleucine.

Our results highlight a novel way to synthesize isoleucine based on the common metabolites glyoxylate and pyruvate. We propose that this novel route could be exploited as an isoleucine production pathway in nature and also in industrial isoleucine production (Lu et al., 2024 [↗](#); Zhang et al., 2024 [↗](#)). This route, unlike the canonical threonine deaminase route, and O-succinylhomoserine cleavage route, is not linked to other amino acid biosynthesis routes, and requires fewer enzymatic steps from central metabolic intermediates. When isoleucine is made via threonine it is dependent on threonine metabolic regulation including attenuation of the *thr* operon (Lynn et al., 1982 [↗](#); Wenk et al., 2025 [↗](#)), allosteric inhibition (Wright & Takahashi, 1977 [↗](#)), and product inhibition (Wright & Takahashi, 1977 [↗](#)). When isoleucine is made via the O-succinylhomoserine cleavage route it is dependent on methionine and cysteine biosynthesis route and needs lower cysteine levels than normal physiological concentrations. Hence, the 2-AHAS route that starts directly from glyoxylate and pyruvate, independent of other amino acid biosynthesis routes (Cotton et al., 2020 [↗](#)) could lead to a more efficient biosynthetic pathway for isoleucine in industrial settings. Furthermore, expression of fewer enzymes results in a lower metabolic burden, which could increase substrate use efficiency and product yield.

We also performed a computational flux balance analysis (FBA) of biomass and product yields for each isoleucine biosynthesis route using both glucose and acetate as feedstocks (Table 1 [↗](#)). This analysis shows that the recursive AHAS route in terms of isoleucine yield is as efficient as any of the isoleucine biosynthesis routes known in nature, including the canonical threonine deaminase routes. The predicted biomass yields both from glucose and acetate are even slightly higher if this route is used than the canonical route, confirming it could also likely be a naturally relevant route for the biosynthesis of this cellular building block.

Another potential application of the findings in the paper could be the use as glyoxylate biosensor. Glyoxylate is a key metabolite in several relevant synthetic pathways, e.g. for alternative synthetic CO<sub>2</sub> fixation routes. A recent work identified and constructed several glyoxylate biosensor strains based on an exhaustive computational analysis of the *E. coli* metabolic network (Orsi et al., 2025 [↗](#)). This analysis was unable to identify the recursive AHAS route due to the fact that this promiscuous activity was previously unreported. Our strain could also be used as a glyoxylate (or 2KB) biosensor based on its low biomass demand and robust growth with minimal genetic disruption (see also Figure 4B [↗](#)).

Overall, our findings bring the number of demonstrated isoleucine biosynthesis routes in *E. coli* MG1655 up to five (Figure 6 [↗](#)) and the total space of isoleucine biosynthesis routes in nature to 10. This highlights the fact that even well-characterized amino acids biosynthesis pathways in a model organism such as *E. coli* are not fully understood, challenging canonical metabolic knowledge.

## Materials and methods

### Strains and plasmids

All strains used in this study are listed in Table 2 [↗](#). The *E. coli* SJJ488 strain based on K-12 MG1655 (Jensen et al., 2016 [↗](#)) was used for the generation of deletion strains.

Table 1.

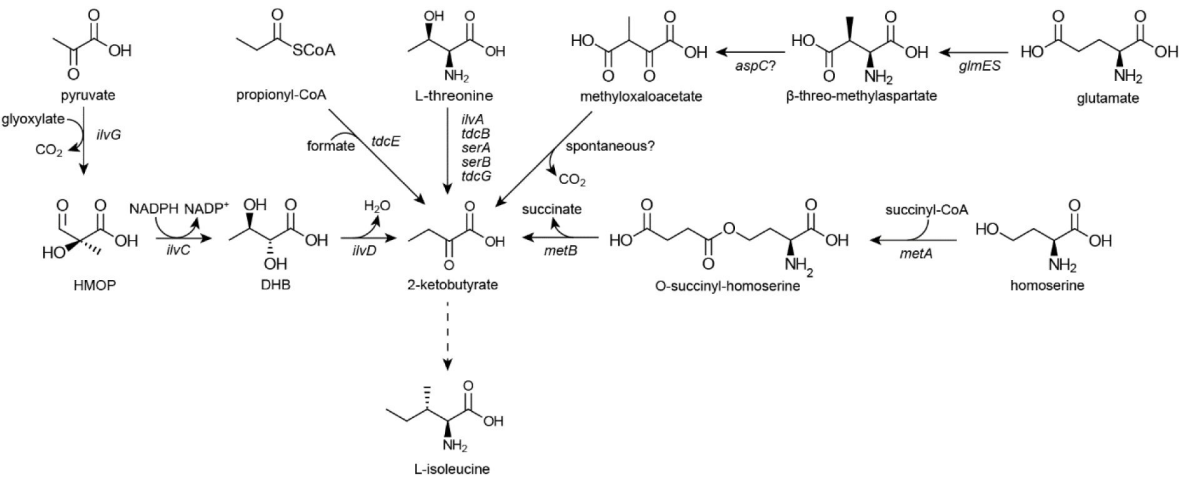
FBA predictions of theoretical biomass and isoleucine yield for the discovered route and other isoleucine biosynthesis routes in *E. coli*.

<sup>a</sup> this route requires an anaerobic environment. NA indicates this route is not applicable as anaerobic growth (without electron acceptor) is not feasible.

| Isoleucine biosynthetic routes                              | Biomass Yield (gCDW/mol glucose) | Maximum isoleucine yield (mol/mol glucose ) | Biomass Yield (gCDW/mol acetate) | Maximum isoleucine yield (mol/mol acetate ) |
|-------------------------------------------------------------|----------------------------------|---------------------------------------------|----------------------------------|---------------------------------------------|
| threonine deaminase (e.g. <i>ilvA</i> )                     | 90.26                            | 0.75                                        | 23.35                            | 0.20                                        |
| recursive AHAS (acetohydroxyacid synthase II, <i>ilvG</i> ) | 90.46                            | 0.75                                        | 23.37                            | 0.20                                        |
| <i>O</i> -succinylhomoserine cleavage ( <i>metB</i> )       | 90.29                            | 0.76                                        | 23.35                            | 0.20                                        |
| 2KB-formate lyase ( <i>tdcE</i> ) <sup>a</sup>              | 18.04                            | 0.80                                        | NA                               | NA                                          |
| glutamate mutase ( <i>glmES</i> ) <sup>a</sup>              | 18.11                            | 0.44                                        | NA                               | NA                                          |

Figure 6.

Summary of the demonstrated isoleucine biosynthesis routes in *E. coli*.



| Strain name         | Genotype                                                                                               | Description                                                                                                               | Source                |
|---------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------|
| SIJ488              | <i>E. coli</i> K-12 MG1655<br>Tn7:: pAra-exo-beta-gam; pRha-FLP; xylSpm-IscI                           | MG1655 derivative with genome integrated recombinase and flippase genes                                                   | (Jensen et al., 2016) |
| Δ5                  | SIJ488 $\Delta ilvA \Delta tdcB \Delta sdaA \Delta sdaB \Delta tdcG::kan$                              | Threonine deaminase and serine deaminase deletion strain                                                                  | (Cotton et al., 2020) |
| IMaux               | SIJ488 $\Delta ilvA \Delta tdcB \Delta sdaA \Delta sdaB \Delta tdcG \Delta metB::kan$                  | Δ5 with O-succinylhomoserine lyase deletion                                                                               | this study            |
| IMaux $\Delta ilvG$ | SIJ488 $\Delta ilvA \Delta tdcB \Delta sdaA \Delta sdaB \Delta tdcG \Delta metB::kan \Delta ilvG::cam$ | Δ5 with O-succinylhomoserine lyase and AHAS-II deletion                                                                   | this study            |
|                     |                                                                                                        |                                                                                                                           |                       |
| Plasmid name        | pCym46-ilvGM                                                                                           | cumate-inducible SEVA vector with spectinomycin resistance and p15A ori for the expression of <i>ilvG</i> and <i>ilvM</i> | this study            |

**Table 2.**

**Strains and plasmid used in this study.**

SIJ488 is engineered to carry the gene deletion machinery in its genome (inducible recombinase and flippase). All gene deletions were carried out by successive rounds of  $\lambda$ -Red recombineering using chloramphenicol or kanamycin (pKD3/pKD4) (Datsenko & Wanner, 2000 [DOI](#)) as described by (Baba et al., 2006 [DOI](#)). All plasmids are listed in **Table 2** [DOI](#) and were generated using a Golden Gate protocol as reported previously (Batianis et al., 2020 [DOI](#)).

## Cultivation conditions

LB medium (1% NaCl, 1% tryptone, 0.5% yeast extract) was used for strain maintenance, generation of deletion strains, and cloning. Antibiotics were used at the following concentrations: chloramphenicol, 25  $\mu\text{g/mL}$ ; kanamycin, 50  $\mu\text{g/mL}$ ; spectinomycin, 100  $\mu\text{g/mL}$ ; apramycin, 50  $\mu\text{g/mL}$ . M9 minimal medium was used for growth experiments (50 mM  $\text{Na}_2\text{HPO}_4$ , 20 mM  $\text{KH}_2\text{PO}_4$ , 1 mM NaCl, 20 mM  $\text{NH}_4\text{Cl}$ , 2 mM  $\text{MgSO}_4$ , 100  $\mu\text{M}$   $\text{CaCl}_2$ , 134  $\mu\text{M}$  EDTA, 13  $\mu\text{M}$   $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ , 6.2  $\mu\text{M}$   $\text{ZnCl}_2$ , 0.76  $\mu\text{M}$   $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ , 0.42  $\mu\text{M}$   $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$ , 1.62  $\mu\text{M}$   $\text{H}_3\text{BO}_3$ , 0.081  $\mu\text{M}$ ,  $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ ) with carbon sources added to the concentrations specified in the text and figures. For growth experiments, overnight cultures were incubated in 3 mL LB medium with appropriate antibiotics. Before inoculation of the experiment, cultures were harvested and washed three times with a saline solution to remove residual carbon sources. Plate reader experiments were inoculated with a starting OD<sub>600</sub> of 0.02 using the washed culture. Plate reader experiments were carried out in 96-well microtiter plates (Nunc Delta Surface, Thermo Fisher Scientific). Each well contained 150  $\mu\text{L}$  of cell culture covered with 50  $\mu\text{L}$  mineral oil (Sigma-Aldrich, Taufkirchen, Germany), to avoid evaporation. A Synergy H1 plate reader (Biotek) was used for incubation (37 °C), shaking, and OD<sub>600</sub> measurements. Three cycles of four shaking phases, each of 1 min were used (1. linear shaking at an amplitude of 3 mm, 2. orbital shaking at an amplitude of 3 mm, 3. linear shaking at an amplitude of 2 mm, and 4. orbital shaking at an amplitude of 2 mm). Optical density (OD 600 nm) was measured after each round of shaking (~12.5 min). Plate reader OD measurements were converted to cuvette values according to the formula  $\text{OD}_{\text{cuvette}} = \text{OD}_{\text{plate}}/0.23$ . Growth curves were processed in python (<https://github.com/he-hai/growth2fig> [DOI](#)) and represent averages of technical triplicate measurements.

## Whole-genome sequencing

Genomic DNA was extracted using the DNeasy Blood and Tissue kit (QIAGEN) following the manufacturer's instructions. Genome sequencing was performed by Plasmidsaurus (London, United Kingdom) using the Oxford Nanopore Technology sequencing platform. Analysis of the sequencing data was performed using the breseq pipeline (Barrick et al., 2014 [DOI](#)) and the SIJ488 reference genome sequence, which is derived from the MG1655 (NC\_000913, GenBank). The mutated *ilvG* sequence is provided in the Supplementary file 1.

## Isotopic labeling of proteinogenic amino acids

Strains for isotopic labeling were grown in M9 supplemented with 1 mM methionine using  $^{13}\text{C}$  labeled glucose and either unlabeled 2KB or unlabeled glyoxylate. The equivalent of 1 mL of OD 1 was harvested, pelleted by centrifugation, and washed twice with sterile distilled water to eliminate leftover traces of medium. The samples were stored at -20 °C and eventually prepared, derivatized and analyzed by GC-MS as previously described (Donati et al., 2023 [DOI](#)). We considered for further analysis fragments listed in (Long & Antoniewicz, 2019 [DOI](#)) Table 3, in particular Ile<sub>274</sub>. Raw chromatographic data was integrated using SmartPeak (Kutuzova et al., 2020 [DOI](#)).

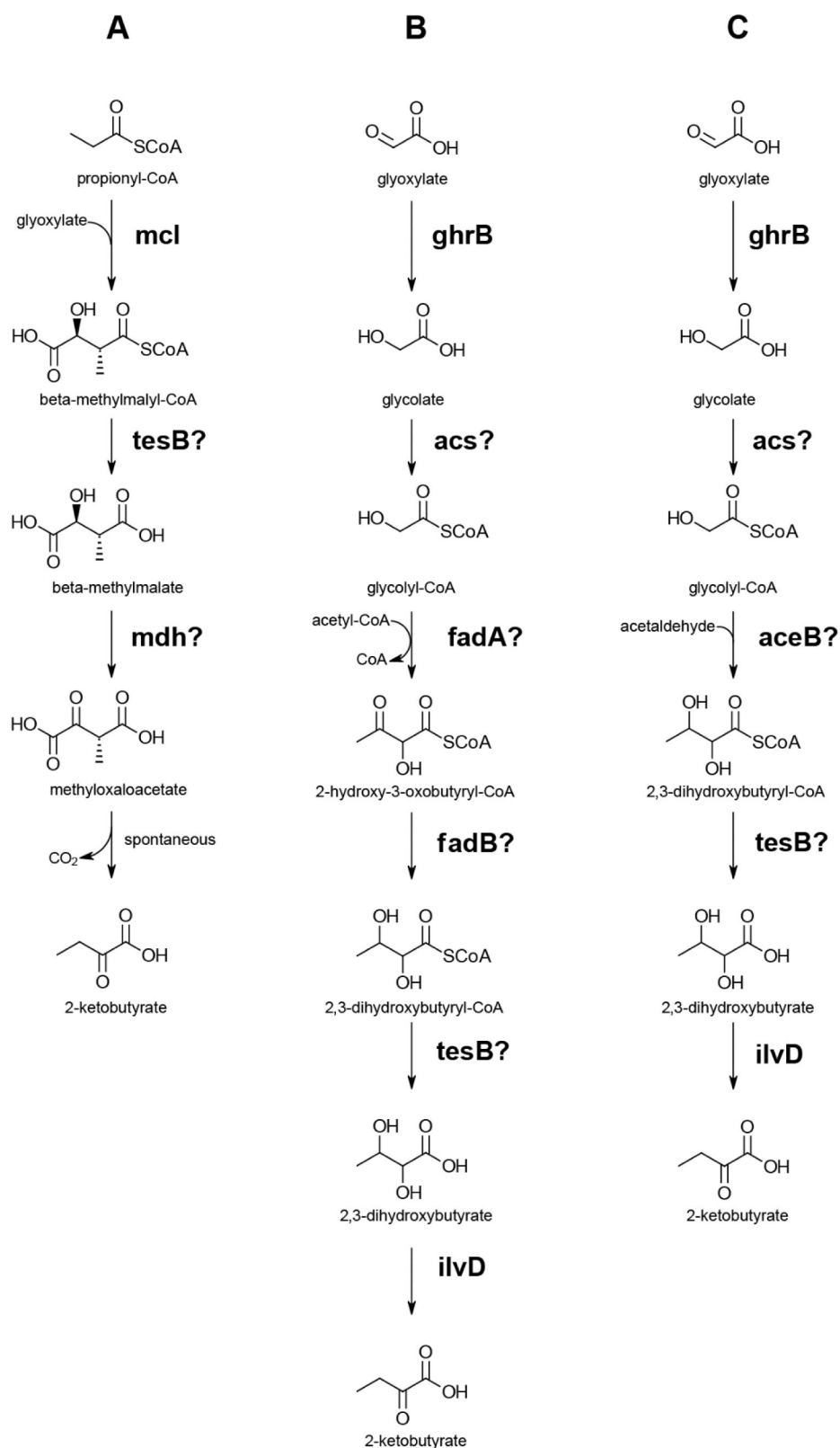
Processed data was further corrected for the natural abundance of isotopes in the derivatization agents used for GCMS analysis (Wahl et al., 2004 [DOI](#)). The data is provided as a Supplementary file 2.

## Flux Balance Analysis

FBA calculations were performed with a custom script on cobrapy version 0.26.3 (Ebrahim et al., 2013 [↗](#)), numpy 2.1.1 and pandas 2.2.2. iML1515 was used as *E. coli* genome scale metabolic model for all calculations (Monk et al., 2017 [↗](#)). The model was modified by changing the membrane transhydrogenase reaction THD2pp to translocate only one proton, and setting non-growth associated maintenance (ATPM) to 0. Anaerobic reactions PFL, POR5, and OBTFL were also set to zero unless specifically relevant. Oxygen import into the cell (EX\_o2\_e) was set to zero when modeling anaerobic growth. The promiscuous isoleucine biosynthesis reactions that are the object of this paper, along with the glutamate mutase route, are not part of this model and were added for the relevant calculations.

## Supplementary figures





**Figure S1.**

**Summary of hypothetical 2KB biosynthesis pathways.**

Question marks indicate that the indicated reaction has not been demonstrated.

## Acknowledgements

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## Additional information

### Author contributions

VR conceived the work, performed the experimental work, data analysis and wrote the draft manuscript. SD performed the  $^{13}\text{C}$  -labelling analysis. SA and NJC supervised the work and acquired the funding. All authors helped in revising the manuscript.

## Additional files

**Supplemental file 1.** [↗](#)

**Supplemental file 2.** [↗](#)

## References

- Abramsky T., Rowland L. P., Shemin D (1962) **The Formation of Isoleucine from  $\beta$ -Methylaspartic Acid in *Escherichia coli* W** *Journal of Biological Chemistry* **237**:C265 [https://doi.org/10.1016/S0021-9258\(18\)81400-4](https://doi.org/10.1016/S0021-9258(18)81400-4) | [Google Scholar](#)
- Andersen D. C., Swartz J., Ryll T., Lin N., Snedecor B (2001) **Metabolic oscillations in an *E. coli* fermentation** *Biotechnology and Bioengineering* **75**:212–218 <https://doi.org/10.1002/bit.10018> | [Google Scholar](#)
- Baba T., Ara T., Hasegawa M., Takai Y., Okumura Y., Baba M., Datsenko K. A., Tomita M., Wanner B. L., Mori H (2006) **Construction of *Escherichia coli* K-12 in-frame, single-gene knockout mutants: The Keio collection** *Molecular Systems Biology* **2** <https://doi.org/10.1038/msb4100050> | [Google Scholar](#)
- Barrick J. E., Colburn G., Deatherage D. E., Traverse C. C., Strand M. D., Borges J. J., Knoester D. B., Reba A., Meyer A. G (2014) **Identifying structural variation in haploid microbial genomes from short- read resequencing data using breseq** *BMC Genomics* **15**:1039 <https://doi.org/10.1186/1471-2164-15-1039> | [Google Scholar](#)
- Batianis C., Damalas S. G., Martin-pascual M., Lorenzo V. D (2020) **SEVA 3.1: Enabling interoperability of DNA assembly among the SEVA, BioBricks and Type IIS restriction enzyme standards** *Microbial Biotechnology* **13**:1793–1806 <https://doi.org/10.1111/1751-7915.13609> | [Google Scholar](#)
- Bisswanger H (1981) **Substrate specificity of the pyruvate dehydrogenase complex from *Escherichia coli*** *Journal of Biological Chemistry* **256**:815–822 [https://doi.org/10.1016/S0021-9258\(19\)70050-7](https://doi.org/10.1016/S0021-9258(19)70050-7) | [Google Scholar](#)
- Burns D. M., Burger M. J., Beacham I. R (1995) **Silent genes in bacteria: The previously designated ‘cryptic’ *ilvHI* locus of ‘*Salmonella typhimurium* LT2’ is active in natural isolates** *FEMS Microbiology Letters* **131**:167–172 <https://doi.org/10.1111/j.1574-6968.1995.tb07772.x> | [Google Scholar](#)
- Claassens N. J., Scarinci G., Fischer A., Flamholz A. I., Newell W., Frielingsdorf S., Lenz O., Bar-Even A (2020) **Phosphoglycolate salvage in a chemolithoautotroph using the Calvin cycle** *Proceedings of the National Academy of Sciences* **117**:22452–22461 <https://doi.org/10.1073/pnas.2012288117> | [Google Scholar](#)
- Cotton C. A. R., Bernhardsgrütter I., He H., Burgener S., Schulz L., Paczia N., Dronsella B., Erban A., Toman S., Dempfle M., De Maria A., Kopka J., Lindner S. N., Erb T. J., Bar-Even A. (2020) **Underground isoleucine biosynthesis pathways in *e. coli*** *eLife* **9**:1–25 <https://doi.org/10.7554/ELIFE.54207> | [Google Scholar](#)
- Daniel J., Dondon L., Danchin A (1983) **2-ketobutyrate: A putative alarmone of *Escherichia coli*** *Molecular and General Genetics MGG* **190**:452–458 <https://doi.org/10.1007/BF00331076> | [Google Scholar](#)
- D’Ari R., Casadesús J (1998) **Underground metabolism** *BioEssays* **20**:181–186 [https://doi.org/10.1002/\(SICI\)1521-1878\(199802\)20:2<181::AID-BIES10>3.0.CO;2-0](https://doi.org/10.1002/(SICI)1521-1878(199802)20:2<181::AID-BIES10>3.0.CO;2-0) | [Google Scholar](#)

- Datsenko K. A., Wanner B. L (2000) **One-step inactivation of chromosomal genes in Escherichia coli K-12 using PCR products** *Proceedings of the National Academy of Sciences of the United States of America* **97**:6640–6645 <https://doi.org/10.1073/pnas.120163297> | Google Scholar
- Donati S., Mattanovich M., Hjort P., Jacobsen S. A. B., Blomquist S. D., Mangaard D., Gurdo N., Pastor F. P., Maury J., Hanke R., Herrgård M. J., Wulff T., Jakočiūnas T., Nielsen L. K., McCloskey D (2023) **An automated workflow for multi-omics screening of microbial model organisms** *Npj Systems Biology and Applications* **9**:14 <https://doi.org/10.1038/s41540-023-00277-6> | Google Scholar
- Ebrahim A., Lerman J. A., Palsson B. O., Hyduke D. R (2013) **COBRapy: COstraints-Based Reconstruction and Analysis for Python** *BMC Systems Biology* **7**:74 <https://doi.org/10.1186/1752-0509-7-74> | Google Scholar
- Eoyang L., Silverman P. M (1984) **Purification and subunit composition of acetohydroxyacid synthase I from Escherichia coli K-12** *Journal of Bacteriology* **157**:184–189 <https://doi.org/10.1128/jb.157.1.184-189.1984> | Google Scholar
- Fang H., Li D., Kang J., Jiang P., Sun J., Zhang D (2018) **Metabolic engineering of Escherichia coli for de novo biosynthesis of vitamin B12** *Nature Communications* **9**:4917 <https://doi.org/10.1038/s41467-018-07412-6> | Google Scholar
- Gollop N., Damri B., Barak Z., Chipman D. M (1989) **Kinetics and mechanism of acetohydroxy acid synthase isozyme III from Escherichia coli** *Biochemistry* **28**:6310–6317 <https://doi.org/10.1021/bi00441a024> | Google Scholar
- Gray J. E., Patin D. W., Calhoun D. H (1981) **Identification of the protein products of the rrnC, ilv, rho region of the Escherichia coli K-12 chromosome** *Molecular and General Genetics MGG* **183**:428–436 <https://doi.org/10.1007/BF00268761> | Google Scholar
- Hill M. C., Pang S. S., Duggleby G. R (1997) **Purification of Escherichia coli acetohydroxyacid synthase isoenzyme II and reconstitution of active enzyme from its individual pure subunits** *Biochemical Journal* **327**:891–898 <https://doi.org/10.1042/bj3270891> | Google Scholar
- Jensen S. I., Lennen R. M., Herrgård M. J., Nielsen A. T (2016) **Seven gene deletions in seven days: Fast generation of Escherichia coli strains tolerant to acetate and osmotic stress** *Scientific Reports* **5**:17874 <https://doi.org/10.1038/srep17874> | Google Scholar
- Khersonsky O., Tawfik D. S (2010) **Enzyme Promiscuity: A Mechanistic and Evolutionary Perspective** *Annual Review of Biochemistry* **79**:471–505 <https://doi.org/10.1146/annurev-biochem-030409-143718> | Google Scholar
- Kim J., Kershner J. P., Novikov Y., Shoemaker R. K., Copley S. D (2010) **Three serendipitous pathways in E. coli can bypass a block in pyridoxal-5'-phosphate synthesis** *Molecular Systems Biology* **6**:436 <https://doi.org/10.1038/msb.2010.88> | Google Scholar
- Kronen M., Berg I. A (2015) **Mesaconase/Fumarase FumD in Escherichia coli O157:H7 and Promiscuity of Escherichia coli Class I Fumarases FumA and FumB** *PLOS One* **10**:e0145098 <https://doi.org/10.1371/journal.pone.0145098> | Google Scholar
- Krüsemann J. L., Lindner S. N., Dempfle M., Widmer J., Arrivault S., Debacker M., He H., Kubis A., Chayot R., Anissimova M., Marlière P., Cotton C. A. R., Bar-Even A (2018) **Artificial pathway**

**emergence in central metabolism from three recursive phosphoketolase reactions** *The FEBS Journal* **285**:4367–4377 <https://doi.org/10.1111/febs.14682> | [Google Scholar](#)

Kubala G., Martell A. E (1981) **Nuclear magnetic resonance investigation of the spontaneous decarboxylation of 2-oxalopropionic acid. 2. Species in solution** *Journal of the American Chemical Society* **103**:7609–7615 <https://doi.org/10.1021/ja00415a032> | [Google Scholar](#)

Kutuzova S., Colaianni P., Röst H., Sachsenberg T., Alka O., Kohlbacher O., Burla B., Torta F., Schrübbers L., Kristensen M., Nielsen L., Herrgård M. J., McCloskey D (2020) **SmartPeak Automates Targeted and Quantitative Metabolomics Data Processing** *Analytical Chemistry* **92**:15968–15974 <https://doi.org/10.1021/acs.analchem.0c03421> | [Google Scholar](#)

LaRossa R. A., Van Dyk T. K., Smulski D. R. (1987) **Toxic accumulation of alpha-ketobutyrate caused by inhibition of the branched-chain amino acid biosynthetic enzyme acetolactate synthase in Salmonella typhimurium** *Journal of Bacteriology* **169**:1372–1378 <https://doi.org/10.1128/jb.169.4.1372-1378.1987> | [Google Scholar](#)

Lawther R. P., Calhoun D. H., Adams C. W., Hauser C. A., Gray J., Hatfield G. W (1981) **Molecular basis of valine resistance in Escherichia coli K-12** *Proceedings of the National Academy of Sciences* **78**:922–925 <https://doi.org/10.1073/pnas.78.2.922> | [Google Scholar](#)

Lawther R. P., Calhoun D. H., Gray J., Adams C. W., Hauser C. A., Hatfield G. W (1982) **DNA sequence fine-structure analysis of ilvG (IlvG+) mutations of Escherichia coli K-12** *Journal of Bacteriology* **149**:294–298 <https://doi.org/10.1128/jb.149.1.294-298.1982> | [Google Scholar](#)

Long C. P., Antoniewicz M. R (2019) **High-resolution 13C metabolic flux analysis** *Nature Protocols* **14**:2856–2877 <https://doi.org/10.1038/s41596-019-0204-0> | [Google Scholar](#)

Lu N., Wei M., Yang X., Li Y., Sun H., Yan Q., Zhang H., He J., Ma J., Xia M., Zhang C (2024) **Growth-coupled production of L-isoleucine in Escherichia coli via metabolic engineering** *Metabolic Engineering* **86**:181–193 <https://doi.org/10.1016/j.ymben.2024.10.004> | [Google Scholar](#)

Lynn S. P., Gardner J. F., Reznikoff W. S (1982) **Attenuation regulation in the thr operon of Escherichia coli K-12: Molecular cloning and transcription of the controlling region** *Journal of Bacteriology* **152**:363–371 <https://doi.org/10.1128/jb.152.1.363-371.1982> | [Google Scholar](#)

Marcheschi R. J., Li H., Zhang K., Noey E. L., Kim S., Chaubey A., Houk K. N., Liao J. C (2012) **A Synthetic Recursive “+1” Pathway for Carbon Chain Elongation** *ACS Chemical Biology* **7**:689–697 <https://doi.org/10.1021/cb200313e> | [Google Scholar](#)

Monk J. M., Lloyd C. J., Brunk E., Mih N., Sastry A., King Z., Takeuchi R., Nomura W., Zhang Z., Mori H., Feist A. M., Palsson B. O (2017) **iML1515, a knowledgebase that computes Escherichia coli traits** *Nature Biotechnology* **35**:904–908 <https://doi.org/10.1038/nbt.3956> | [Google Scholar](#)

Neidhardt F (1987) **Escherichia coli and Salmonella typhimurium: Cellular and molecular biology** American Society for Microbiology [Google Scholar](#)

- Noda-Garcia L., Liebermeister W., Tawfik D. S (2018) **Metabolite-Enzyme Coevolution: From Single Enzymes to Metabolic Pathways and Networks** *Annual Review of Biochemistry* **87**:187–216 <https://doi.org/10.1146/annurev-biochem-062917-012023> | Google Scholar
- Noor E., Eden E., Milo R., Alon U (2010) **Central Carbon Metabolism as a Minimal Biochemical Walk between Precursors for Biomass and Energy** *Molecular Cell* **39**:809–820 <https://doi.org/10.1016/j.molcel.2010.08.031> | Google Scholar
- Oh M.-K., Liao J. C (2000) **Gene Expression Profiling by DNA Microarrays and Metabolic Fluxes in Escherichiacoli** *Biotechnology Progress* **16**:278–286 <https://doi.org/10.1021/bp000002n> | Google Scholar
- Orsi E., Claassens N. J., Nikel P. I., Lindner S. N (2021) **Growth-coupled selection of synthetic modules to accelerate cell factory development** *Nature Communications* **12**:1–5 <https://doi.org/10.1038/s41467-021-25665-6> | Google Scholar
- Orsi E., Schulz-Mirbach H., Cotton C. A. R., Satanowski A., Petri H. M., Arnold S. L., Grabarczyk N., Verbakel R., Jensen K. S., Donati S., Paczia N., Glatter T., Küffner A. M., Chotel T., Schillmüller F., De Maria A., He H., Lindner S. N., Noor E., ..., Nikel P. I. (2025) **Computation-aided designs enable developing auxotrophic metabolic sensors for wide-range glyoxylate and glycolate detection** *Nature Communications* **16**:2168 <https://doi.org/10.1038/s41467-025-57407-3> | Google Scholar
- Phillips A. T., Nuss J. I., Moosic J., Foshay C (1972) **Alternate Pathway for Isoleucine Biosynthesis in Escherichia coli** *Journal of Bacteriology* **109**:714–719 <https://doi.org/10.1128/jb.109.2.714-719.1972> | Google Scholar
- Pontrelli S., Fricke R. C. B., Teoh S. T., Laviña W. A., Putri S. P., Fitz-Gibbon S., Chung M., Pellegrini M., Fukusaki E., Liao J. C (2018) **Metabolic repair through emergence of new pathways in Escherichia coli** *Nature Chemical Biology* **14**:1005–1009 <https://doi.org/10.1038/s41589-018-0149-6> | Google Scholar
- Salmon K. A., Yang C.-R., Hatfield G. W (2006) **Biosynthesis and Regulation of the Branched-Chain Amino Acids** *EcoSal Plus* **2** <https://doi.org/10.1128/ecosalplus.3.6.1.5> | Google Scholar
- Schulz-Mirbach H., Dronsella B., Erb T. J (2025) **Escherichia coli selection strains for growth-coupled metabolic engineering** *Trends in Biotechnology* <https://doi.org/10.1016/j.tibtech.2025.06.015> | Google Scholar
- Steinmetz A., Vyazmensky M., Meyer D., Barak Z., Golbik R., Chipman D. M., Tittmann K (2010) **Valine 375 and Phenylalanine 109 Confer Affinity and Specificity for Pyruvate as Donor Substrate in Acetohydroxy Acid Synthase Isozyme II from Escherichia coli** *Biochemistry* **49**:5188–5199 <https://doi.org/10.1021/bi100555q> | Google Scholar
- Timm S., Sun H., Hagemann M., Huang W., Fernie A. R (2025) **An old dog with new tricks – the value of photorespiration as a central metabolic hub with implications for environmental acclimation.** *Plant Physiology* **kiaf 258** <https://doi.org/10.1093/plphys/kiaf258> | Google Scholar
- Traxler M. F., Summers S. M., Nguyen H.-T., Zacharia V. M., Hightower G. A., Smith J. T., Conway T (2008) **The global, ppGpp-mediated stringent response to amino acid starvation in**

**Escherichia coli** *Molecular Microbiology* **68**:1128–1148 <https://doi.org/10.1111/j.1365-2958.2008.06229.x> | [Google Scholar](#)

Tsai C. S (1967) **Spontaneous decarboxylation of oxalacetic acid** *Canadian Journal of Chemistry* **45**:873–880 <https://doi.org/10.1139/v67-145> | [Google Scholar](#)

Umbarger H. E (1978) **Amino Acid Biosynthesis and its Regulation** *Annual Review of Biochemistry* **47**:533–606 <https://doi.org/10.1146/annurev.bi.47.070178.002533> | [Google Scholar](#)

Unden G., Strecker A., Kleefeld A., Kim O. B (2016) **C 4 -Dicarboxylate Utilization in Aerobic and Anaerobic Growth** *EcoSal Plus* **7** <https://doi.org/10.1128/ecosalplus.ESP-0021-2015> | [Google Scholar](#)

Vyazmensky M., Steinmetz A., Meyer D., Golbik R., Barak Z., Tittmann K., Chipman D. M (2011) **Significant Catalytic Roles for Glu47 and Gln 110 in All Four of the C–C Bond-Making and -Breaking Steps of the Reactions of Acetohydroxyacid Synthase II** *Biochemistry* **50**:3250–3260 <https://doi.org/10.1021/bi102051h> | [Google Scholar](#)

Wahl S. A., Dauner M., Wiechert W (2004) **New tools for mass isotopomer data evaluation in 13C flux analysis: Mass isotope correction, data consistency checking, and precursor relationships** *Biotechnology and Bioengineering* **85**:259–268 <https://doi.org/10.1002/bit.10909> | [Google Scholar](#)

Wenk S., Claassens N. J., Lindner S. N (2022) **Synthetic metabolism approaches: A valuable resource for systems biology** *Current Opinion in Systems Biology* **30**:100417 <https://doi.org/10.1016/j.coisb.2022.100417> | [Google Scholar](#)

Wenk S., Rainaldi V., Schann K., He H., Bouzon M., Döring V., Lindner S. N., Bar-Even A (2025) **Evolution-assisted engineering of E. coli enables growth on formic acid at ambient CO2 via the Serine Threonine Cycle** *Metabolic Engineering* **88**:14–24 <https://doi.org/10.1016/j.ymben.2024.10.007> | [Google Scholar](#)

Wright K., Takahashi M (1977) **Fluorescence energy transfer between heterologous active sites of affinity-labeled aspartokinase of Escherichia coli** *Biochemistry* **16**:1548–1554 <https://doi.org/10.1021/bi00627a003> | [Google Scholar](#)

Zhang H., Ye T., Fengmin L., Zhang X., Wang J., Wei X., Neo Y. P., Liu H., Fang H (2024) **Strategies to Enhance l-Isoleucine Synthesis by Modifying the Threonine Metabolism Pathway in Escherichia coli** *ACS Omega* **9**:10276–10285 <https://doi.org/10.1021/acsomega.3c07619> | [Google Scholar](#)

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**Reviewer #1 (Public review):**

As presented in this short report, the focus is to only establish that acetohydroxyacid synthase II can have underground activity to generate 2-ketobutyrate (from glyoxylate and pyruvate). Additionally, the gene that encodes this protein has an inactivating point mutation in the lab strain of *E. coli*. In strains lacking the conventional Ile biosynthesis pathway, this enzyme gets reactivated (after short-term laboratory evolution) and putatively can contribute to producing sufficient 2-ketobutyrate, which can feed into Ile production. This is clearly a very interesting observation and finding, and the paper focuses on this single point.

However, the manuscript as it currently stands is 'minimal', and just barely shows that this reaction/pathway is feasible. There is no characterization of the restored enzyme's activity, rate, or specificity. Additionally, there is no data presented on how much isoleucine can be produced, even at saturating concentrations of glyoxylate or pyruvate. This would greatly benefit from more rigorous characterization of this enzyme's activity and function, as well as better demonstration of how effective this pathway is in generating 2-ketobutyrate (and then its subsequent condensation with pyruvate).

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**Reviewer #2 (Public review):**

Summary:

The manuscript by Rainaldi et al. reports a new sub-pathway for isoleucine biosynthesis by demonstrating the promiscuous activity of the native enzyme acetohydroxyacid synthase II (AHAS II). AHAS-II is primarily known to catalyze the condensation of 2-ketobutyrate (2KB) with pyruvate to form a further downstream intermediate, AHB, in the isoleucine



biosynthesis pathway. However, the catalysis of pyruvate and glyoxylate condensation to produce 2KB via the *ilvG* encoded AHAS II is reported in this manuscript for the first time.

Using an isoleucine/2KB auxotrophic *E. coli* strain, the authors report (i) repair of the inactivating frameshift mutation in the *ilvG* gene, which encodes AHAS-II, supports growth in glyoxylate-supplemented media, (ii) the promiscuity of AHAS-II in glyoxylate and pyruvate condensation, resulting in the formation of isoleucin precursors (2-KB), aiding the biosynthesis of isoleucine, and (iii) comparable efficiency of the recursive AHAS-II route to the canonical routes of isoleucin biosynthesis via computational Flux-based analysis.

**Strengths:**

The authors have used laboratory evolution to uncover a non-canonical metabolic route. The metabolomics and FBA have been used to strengthen the claim.

**Weaknesses:**

While the manuscript proposes an interesting metabolic route for the isoleucine biosynthesis, the data lack key controls, biological replicates, and consistency. The figures and methods are presented inadequately. In the current state, the data fails to support the claims made in the manuscript.

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**Author response:**

We gratefully acknowledge the comments on our manuscript and the time you took to read and understand our work. Nevertheless, it is the opinion of these authors that the evidence provided in the submitted paper is strong and we performed multiple replicates of the experiments. In particular, gene deletion and complementation is the accepted gold standard for studies in physiology. In the isoleucine auxotroph (IMaux) strain carrying an *ilvG* deletion, growth is only possible if *ilvG* is reintroduced on a plasmid and induced. Additionally, isotopic labeling clearly demonstrates the activity of the proposed pathway. Regardless, we agree with the reviewers that the paper and the scientific community would benefit from an *in vitro* characterization of the promiscuity of *IlvG*, so we will perform this experiment and resubmit the paper for further revision, and in this revision also provide more detail on the replicates performed.

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